Journal Scope Drift Report

CWTS Leiden citation clustering · macro level · baseline 2020 → 2026 · run 20260721_122750

38,222	15.6%	3 / 8	0.289
Papers analysed	Overall out-of-scope	High-drift journals	Mean JSD 2026

Summary

This report looks at 8 Frontiers journals and asks a simple question: is the type of research each one publishes drifting away from where it started in 2020? Across all of them, about 1 in 6 papers (15.6%) now falls outside the journal's core subject areas. Three journals — Materials, Neurorobotics and Environmental Science — have moved the furthest from their original focus. A page of plain-language commentary and supporting detail follows for each journal.

Scope Drift & OOS Summary by Journal

Journal	Papers	OOS	OOS %	JSD 2026	Drift
Fr. Materials	3,123	453	14.5%	0.379	High
Fr. Neurorobotics	1,037	203	19.6%	0.378	High
Fr. Environmental Science	7,093	1,367	19.3%	0.343	High
Fr. Robotics and AI	1,791	277	15.5%	0.314	Moderate
Fr. Earth Science	7,870	639	8.1%	0.280	Moderate
Fr. Chemistry	6,003	953	15.9%	0.239	Low
Fr. Surgery	5,885	1,052	17.9%	0.232	Low
Fr. Aging Neuroscience	5,420	1,021	18.8%	0.145	Low

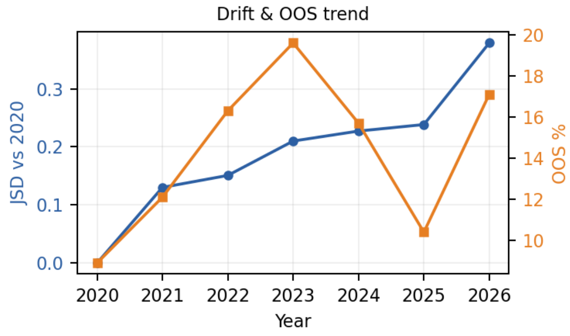
Frontiers in Materials

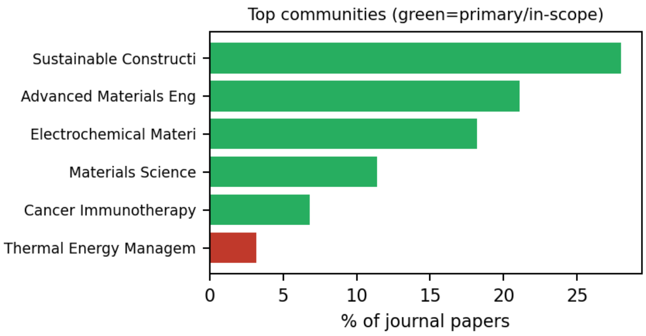
High

Materials shows the clearest signs of scope movement in this review. Compared with where the journal stood in 2020, the mix of research it publishes has shifted noticeably — a change of this size (JSD 0.38) means the papers coming in today look meaningfully different from the ones that defined the journal a few years ago.

Right now, about 14% of Materials's papers (453 out of 3,123) fall outside its core subject clusters — mostly work related to Thermal Energy Management, Dental Health. This out-of-scope share has risen since 2020. At the same time, the journal is concentrating more tightly around a smaller set of themes rather than spreading out. Its largest single topic area today is “Sustainable Construction Materials.”

Papers	Out-of-scope	JSD 2026	Entropy Δ	Primary clusters	Primary coverage
3,123	**453 (14.5%)**	**0.379**	**-0.228**	**5**	**80.0%**





Community composition

Community	Status	Papers	Share of journal
Sustainable Construction Materials	In-scope	875	28.0%
Advanced Materials Engineering	In-scope	660	21.1%
Electrochemical Materials	In-scope	567	18.2%
Materials Science	In-scope	355	11.4%
Cancer Immunotherapy	In-scope	213	6.8%
Thermal Energy Management	Out-of-scope	100	3.2%
Dental Health	Out-of-scope	52	1.7%
Geological Sciences	Out-of-scope	48	1.5%

Example papers

In-scope	Out-of-scope
Prediction of flow stress and microstructure evolution mechanism during thermal tensile process of ZK60 alloy	The PFILSTM model: a crack recognition method based on pyramid features and memory mechanisms
Pitting Judgment Model Based on Machine Learning and Feature Optimization Methods	Magnetic fluid sealing status estimation based on acoustic emission monitoring
In situ Resource Utilization and Reconfiguration of Soils Into Construction Materials for the Additive Manufacturing of Buildings	Sub-severe hail: the missing piece in assessing asphalt shingle risk in North America
Global trends in the research of biodegradable biomedical magnesium-based materials: a bibliometric analysis	Deep learning-based image classification for microstructural analysis in computational materials science

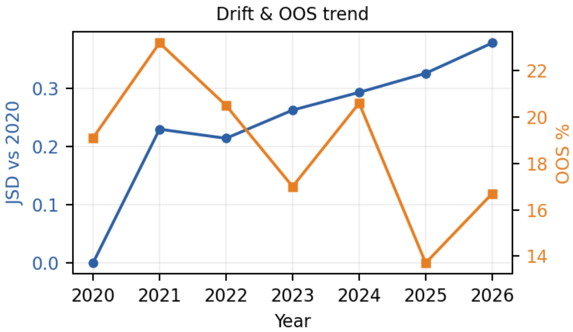
Frontiers in Neurorobotics

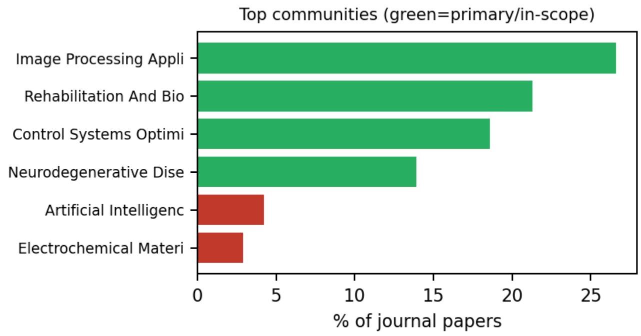
High

Neurorobotics shows the clearest signs of scope movement in this review. Compared with where the journal stood in 2020, the mix of research it publishes has shifted noticeably — a change of this size (JSD 0.38) means the papers coming in today look meaningfully different from the ones that defined the journal a few years ago.

Right now, about 20% of Neurorobotics's papers (203 out of 1,037) fall outside its core subject clusters — mostly work related to Artificial Intelligence Applications, Electrochemical Materials. This out-of-scope share has fallen since 2020. At the same time, the journal is concentrating more tightly around a smaller set of themes rather than spreading out. Its largest single topic area today is “Image Processing Applications.”

Papers	Out-of-scope	JSD 2026	Entropy Δ	Primary clusters	Primary coverage
1,037	**203 (19.6%)**	**0.378**	**-0.4157**	**4**	**80.0%**





Community composition

Community	Status	Papers	Share of journal
Image Processing Applications	In-scope	276	26.6%
Rehabilitation And Biomechanics	In-scope	221	21.3%
Control Systems Optimization	In-scope	193	18.6%
Neurodegenerative Diseases	In-scope	144	13.9%
Artificial Intelligence Applications	Out-of-scope	44	4.2%
Electrochemical Materials	Out-of-scope	30	2.9%
Sustainable Development	Out-of-scope	27	2.6%
Fault Diagnosis	Out-of-scope	21	2.0%

Example papers

In-scope	Out-of-scope
Noisy Dueling Double Deep Q-Network Algorithm for Autonomous Underwater Vehicle Path Planning	The use of deep learning technology in dance movement generation
Neuro-motor Controlled Wearable Augmentations: Current Research and Emerging Trends	Optimal UAV's Deployment and Transmit Power Design for Two Users Uplink NOMA Systems
4D Trajectory Lightweight Prediction Algorithm Based on Knowledge Distillation Technique	The SocialAI School: A framework leveraging Developmental Psychology Towards Artificial Socio-Cultural Agents
A method for the estimation of a motor unit innervation zone center position evaluated with a computational sEMG model	Editorial: Robust Artificial Intelligence for Neurorobotics

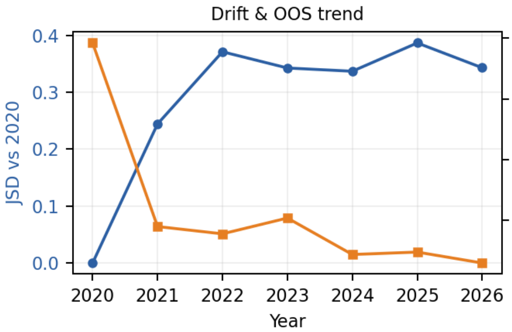
Frontiers in Environmental Science

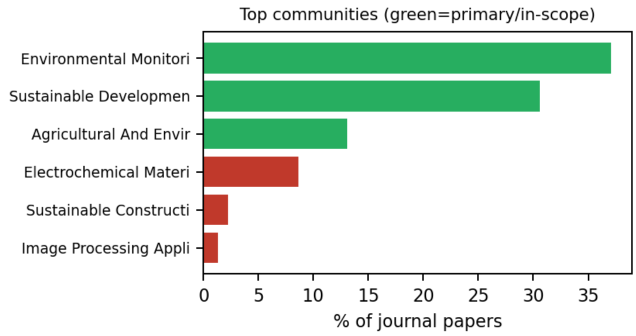
High

Environmental Science shows the clearest signs of scope movement in this review. Compared with where the journal stood in 2020, the mix of research it publishes has shifted noticeably — a change of this size (JSD 0.34) means the papers coming in today look meaningfully different from the ones that defined the journal a few years ago.

Right now, about 19% of Environmental Science's papers (1,367 out of 7,093) fall outside its core subject clusters — mostly work related to Electrochemical Materials, Sustainable Construction Materials. This out-of-scope share has fallen since 2020. The spread of topics it covers has stayed fairly stable. Its largest single topic area today is “Environmental Monitoring.”

Papers	Out-of-scope	JSD 2026	Entropy Δ	Primary clusters	Primary coverage
7,093	**1,367 (19.3%)**	**0.343**	**-0.0436**	**3**	**80.0%**





Community composition

Community	Status	Papers	Share of journal
Environmental Monitoring	In-scope	2,629	37.1%
Sustainable Development	In-scope	2,171	30.6%
Agricultural And Environmental Sciences	In-scope	926	13.1%
Electrochemical Materials	Out-of-scope	608	8.6%
Sustainable Construction Materials	Out-of-scope	156	2.2%
Image Processing Applications	Out-of-scope	91	1.3%
Geological Sciences	Out-of-scope	59	0.8%
Antimicrobial Resistance	Out-of-scope	55	0.8%

Example papers

In-scope	Out-of-scope
Large-scale experimental warming reduces soil faunal biodiversity through peatland drying	Enhancing environmental management through big data: spatial analysis of urban ecological governance and big data development
Assessing Ecological Environmental Quality and Conservation Effectiveness in the World's Largest Urban Green Heart Using the Remote Sensing Ecological Index (RSEI) and Propensity Score Matching (PSM)	Effects of Topographic Variables on Traffic-related Pollutant Concentrations: Comparison of AERMOD and CAL3QHCR Models
Geospatial Environmental Influence on Forest Carbon Sequestration Potential of Tropical Forest Growth in Hainan Island, China	Study on the biodegradation of phenol by Alcaligenes faecalis JH1 immobilized in rice husk biochar

Changes in Urban Gas-Phase Persistent Organic Pollutants During the COVID-19 Lockdown in Barcelona

Evaluation of the factors affecting arsenic distribution using geospatial analysis techniques in Dongting Plain, China

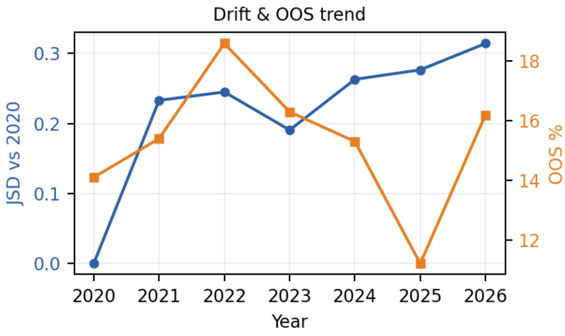
Frontiers in Robotics and AI

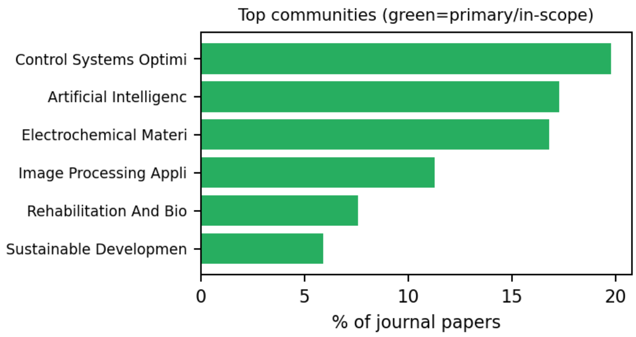
Moderate

Robotics and AI is showing a moderate amount of movement away from its 2020 starting point (JSD 0.31). This isn't dramatic, but the journal's content mix has evolved enough to be worth keeping an eye on.

Right now, about 16% of Robotics and AI's papers (277 out of 1,791) fall outside its core subject clusters — mostly work related to Virtual Reality Education, Space Weather Studies. This out-of-scope share has risen since 2020. The spread of topics it covers has stayed fairly stable. Its largest single topic area today is “Control Systems Optimization.”

Papers	Out-of-scope	JSD 2026	Entropy ∆	Primary clusters	Primary coverage
1,791	**277 (15.5%)**	**0.314**	**+0.0379**	**7**	**80.0%**





Community composition

Community	Status	Papers	Share of journal
Control Systems Optimization	In-scope	354	19.8%
Artificial Intelligence Applications	In-scope	309	17.3%
Electrochemical Materials	In-scope	301	16.8%
Image Processing Applications	In-scope	203	11.3%
Rehabilitation And Biomechanics	In-scope	136	7.6%
Sustainable Development	In-scope	106	5.9%
Neurodegenerative Diseases	In-scope	105	5.9%
Virtual Reality Education	Out-of-scope	24	1.3%

Example papers

In-scope	Out-of-scope
Editorial: Robotic Manipulation and Capture in Space	A robot localization proposal for the RobotAtFactory 4.0: A novel robotics competition within the Industry 4.0 concept
Drive competition underlies effective allostatic orchestration	From the lab to the field with Evolutionary Field Robotics
Trauma-Informed Care Approach in Developing Companion Robots: A Preliminary Observational Study	Rehabilitation assessment of upper limb motor function in stroke patients based on semi-quantitative information
Risk-Aware Model-Based Control	Impact analysis of cooperative perception on the performance of automated driving in unsignalized roundabouts

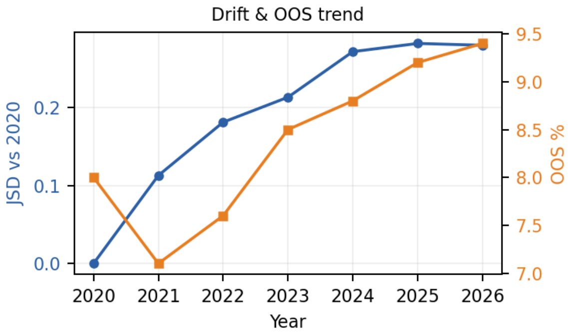
Frontiers in Earth Science

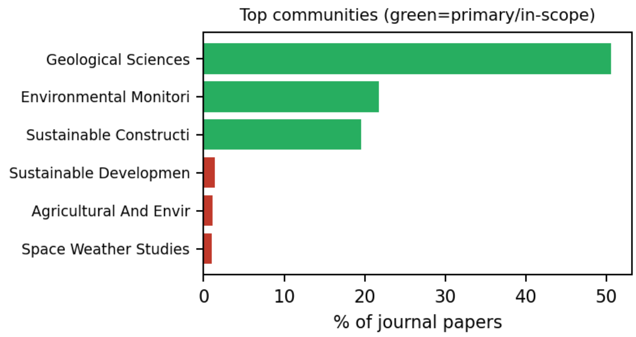
Moderate

Earth Science is showing a moderate amount of movement away from its 2020 starting point (JSD 0.28). This isn't dramatic, but the journal's content mix has evolved enough to be worth keeping an eye on.

Right now, about 8% of Earth Science's papers (639 out of 7,870) fall outside its core subject clusters — mostly work related to Sustainable Development, Agricultural And Environmental Sciences. This out-of-scope share has risen since 2020. The journal is also spreading across a wider range of topics rather than staying tightly focused. Its largest single topic area today is “Geological Sciences.”

Papers	Out-of-scope	JSD 2026	Entropy Δ	Primary clusters	Primary coverage
7,870	**639 (8.1%)**	**0.280**	**+0.1262**	**3**	**80.0%**





Community composition

Community	Status	Papers	Share of journal
Geological Sciences	In-scope	3,981	50.6%
Environmental Monitoring	In-scope	1,716	21.8%
Sustainable Construction Materials	In-scope	1,534	19.5%
Sustainable Development	Out-of-scope	109	1.4%
Agricultural And Environmental Sciences	Out-of-scope	87	1.1%
Space Weather Studies	Out-of-scope	80	1.0%
Image Processing Applications	Out-of-scope	80	1.0%
Electrochemical Materials	Out-of-scope	65	0.8%

Example papers

In-scope	Out-of-scope
An efficient spectral element method for two-dimensional magnetotelluric modeling	Mechanical characteristics analysis of horizontal lifting of subsea pipeline with different burial depths
Stress triggering of the 2022 Lushan–Maerkang earthquake sequence by historical events and its implication for fault stress evolution in eastern Tibet	Evolution of roof treatment processes from blasting to hydraulic fracturing to prevent rockburst in thick-hard roof coal mines in Xinjiang, China
Comparing Greenland Ice Sheet Melt Variability From Different Satellite Passive Microwave Remote Sensing Products Over a Common 5-year Record	On evaluating the possible hazard of soil contamination in arid regions using statistical analysis and GIS techniques
Specific features of marine electrotomographic studies of coastal areas of the gulf of Finland	Editorial: Developments of remote sensing and numerical modeling applications for landslide analysis

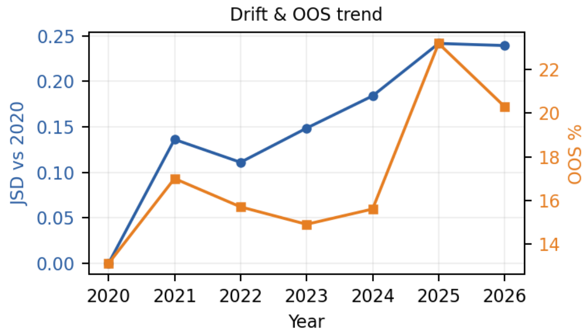
Frontiers in Chemistry

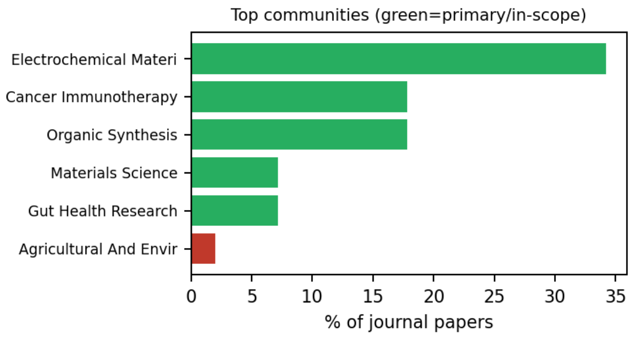
Low

Chemistry remains close to its original scope. Its content mix has changed only slightly since 2020 (JSD 0.24), which suggests the journal is publishing consistently within its intended focus.

Right now, about 16% of Chemistry's papers (953 out of 6,003) fall outside its core subject clusters — mostly work related to Agricultural And Environmental Sciences, COVID-19 Health Impacts. This out-of-scope share has risen since 2020. The journal is also spreading across a wider range of topics rather than staying tightly focused. Its largest single topic area today is “Electrochemical Materials.”

Papers	Out-of-scope	JSD 2026	Entropy Δ	Primary clusters	Primary coverage
6,003	**953 (15.9%)**	**0.239**	**+0.2844**	**5**	**80.0%**





Community composition

Community	Status	Papers	Share of journal
Electrochemical Materials	In-scope	2,051	34.2%
Cancer Immunotherapy	In-scope	1,068	17.8%
Organic Synthesis	In-scope	1,067	17.8%
Materials Science	In-scope	434	7.2%
Gut Health Research	In-scope	430	7.2%
Agricultural And Environmental Sciences	Out-of-scope	123	2.0%
COVID-19 Health Impacts	Out-of-scope	120	2.0%
Antimicrobial Resistance	Out-of-scope	93	1.5%

Example papers

In-scope	Out-of-scope
Fuzzy Logic, Artificial Neural Network, and Adaptive Neuro-Fuzzy Inference Methodology for Soft Computation and Modeling of Ion Sensing Data of a Terpyridyl-Imidazole Based Bifunctional Receptor	Spectral denoising based on Hilbert–Huang transform combined with F-test
Identification, Efficacy, and Stability Evaluation of Succinimide Modification With a High Abundance in the Framework Region of Golimumab	Accelerate the Electrolyte Perturbed-Chain Statistical Associating Fluid Theory–Density Functional Theory Calculation With the Chebyshev Pseudo-Spectral Collocation Method. Part II. Spherical Geometry and Anderson Mixing
Review of Synthesis and Evaluation of Inhibitor Nanomaterials for Oilfield Mineral Scale Control	Recent Advances in Metal/Alloy Nano Coatings for Carbon Nanotubes Based on Electroless Plating

Editorial: Chemical Insights Into the Synthetic Chemistry of Quinazolines and Quinazolinones: Recent Advances

Nanostructures Formed by Custom-Made Peptides Based on Amyloid Peptide Sequences and Their Inhibition by 2-Hydroxynaphthoquinone

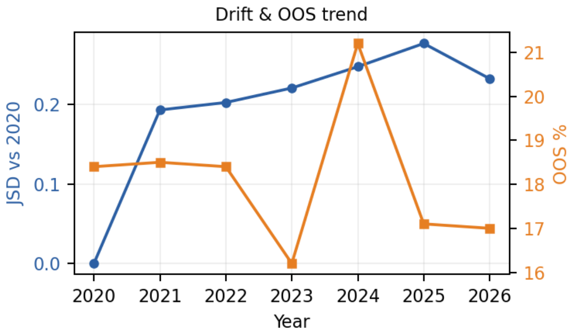
Frontiers in Surgery

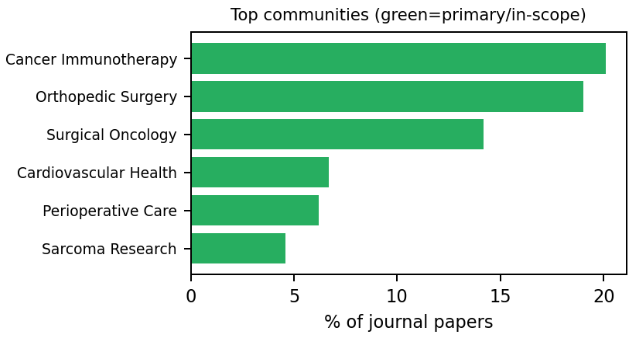
Low

Surgery remains close to its original scope. Its content mix has changed only slightly since 2020 (JSD 0.23), which suggests the journal is publishing consistently within its intended focus.

Right now, about 18% of Surgery's papers (1,052 out of 5,885) fall outside its core subject clusters — mostly work related to peripheral topics. This out-of-scope share has fallen since 2020. At the same time, the journal is concentrating more tightly around a smaller set of themes rather than spreading out. Its largest single topic area today is “Cancer Immunotherapy.”

Papers	Out-of-scope	JSD 2026	Entropy Δ	Primary clusters	Primary coverage
5,885	**1,052 (17.9%)**	**0.232**	**-0.1047**	**10**	**80.0%**





Community composition

Community	Status	Papers	Share of journal
Cancer Immunotherapy	In-scope	1,181	20.1%
Orthopedic Surgery	In-scope	1,119	19.0%
Surgical Oncology	In-scope	834	14.2%
Cardiovascular Health	In-scope	397	6.7%
Perioperative Care	In-scope	362	6.2%
Sarcoma Research	In-scope	272	4.6%
Reproductive Health	In-scope	201	3.4%
Hearing Disorders	In-scope	200	3.4%

Example papers

In-scope	Out-of-scope
Correlation Between Poor Defecation Habits and Postoperative Hemorrhoid Recurrence	The Heterogenous Effect of COVID-19 on Liver Transplantation Activity and Waitlist Mortality in the United States
Case report: Giant cystic ileal gastrointestinal stromal tumor with an atypical intratumoral abscess	Facial nerve reconstruction for flaccid facial paralysis: a systematic review and meta-analysis
Comprehensive clinical analysis of patients with primary malignant tumor of pituitary gland: A population-based study	Case report: Type I diastematomyelia with breast abnormalities and clubfoot
Relationship Between Serum ECP and TIgE Levels and the Risk of Postoperative Recurrence in Patients with Chronic Rhinosinusitis with Nasal Polyps	A survey on nasoalveolar moulding treatment practices at cleft centres across India

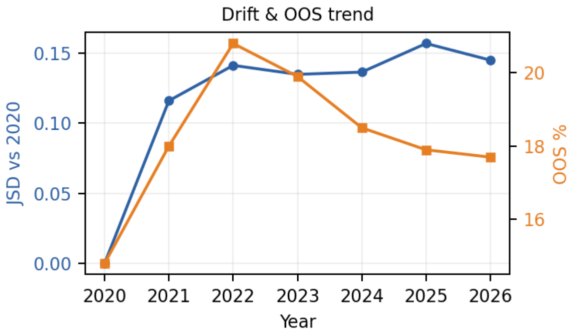
Frontiers in Aging Neuroscience

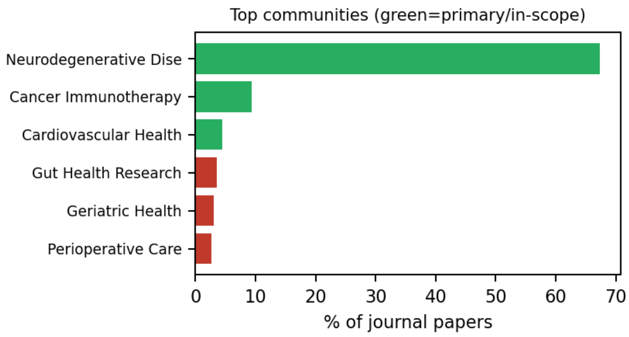
Low

Aging Neuroscience remains close to its original scope. Its content mix has changed only slightly since 2020 (JSD 0.14), which suggests the journal is publishing consistently within its intended focus.

Right now, about 19% of Aging Neuroscience's papers (1,021 out of 5,420) fall outside its core subject clusters — mostly work related to Gut Health Research, Geriatric Health. This out-of-scope share has risen since 2020. The journal is also spreading across a wider range of topics rather than staying tightly focused. Its largest single topic area today is “Neurodegenerative Diseases.”

Papers	Out-of-scope	JSD 2026	Entropy Δ	Primary clusters	Primary coverage
5,420	1,021 (18.8%)	0.145	+0.1983	3	80.0%





Community composition

Community	Status	Papers	Share of journal
Neurodegenerative Diseases	In-scope	3,652	67.4%
Cancer Immunotherapy	In-scope	503	9.3%
Cardiovascular Health	In-scope	244	4.5%
Gut Health Research	Out-of-scope	195	3.6%
Geriatric Health	Out-of-scope	163	3.0%
Perioperative Care	Out-of-scope	143	2.6%
Rehabilitation And Biomechanics	Out-of-scope	134	2.5%
COVID-19 Health Impacts	Out-of-scope	66	1.2%

Example papers

In-scope	Out-of-scope
Electroacupuncture Ameliorates Cognitive Impairment by Inhibiting the JNK Signaling Pathway in a Mouse Model of Alzheimer’s Disease	Baseline CSF ferritin levels were associated with trajectories of depressive symptoms among older people without dementia
Correlation between cerebral small vessel disease and postural instability/gait difficulty subtype in Parkinson’s disease patients	Age and APOE affect L-carnitine system metabolites in the brain in the APOE-TR model
BrainFD: Measuring the Intracranial Brain Volume With Fractal Dimension	Decanoic acid, an MCT dietary component, alleviates cognitive impairment, cellular senescence, and promotes autophagy in accelerated aging and neurotoxic mouse models induced by chronic administration of D-galactose and D-galactose/AlCl3
Various Diseases and Clinical Heterogeneity Are Associated With “Hot Cross Bun”	Cognition and Cerebrovascular Reactivity in Midlife Women With History of Preeclampsia and Placental Evidence of Maternal Vascular Malperfusion